

Hall A Polarimetry – Future Plans

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Hall A Collaboration Meeting
February 10-11, 2021

Outline

- Hall A Polarimetry
- Improvements to Hall A (and C) Compton Polarimeters (HIPPOL)
 - Laser system
 - Electron detector
- Møller polarimeter
 - Tracking detectors
 - Collimator
 - New target position

Hall A Polarimetry

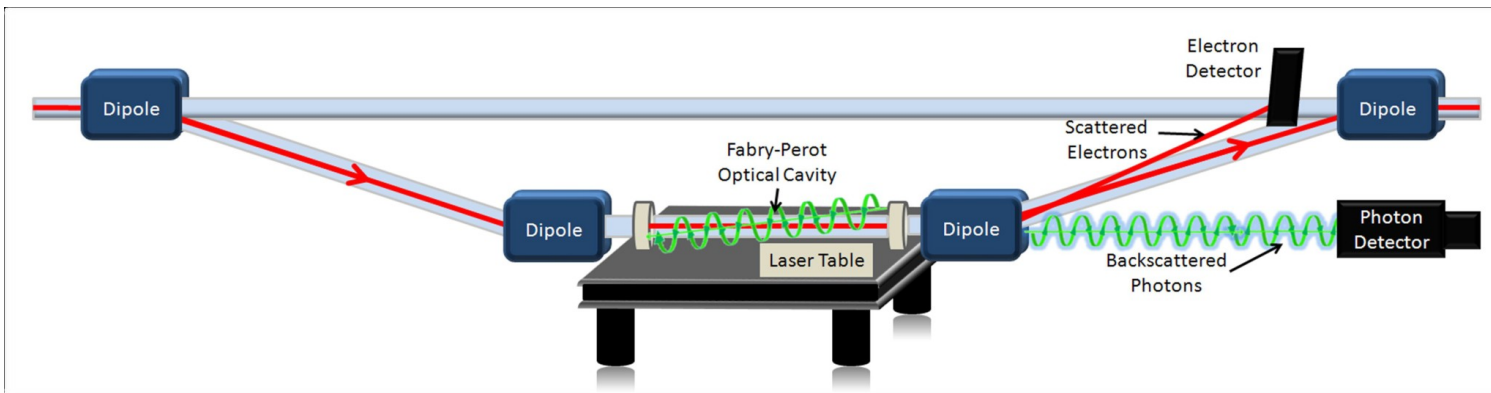
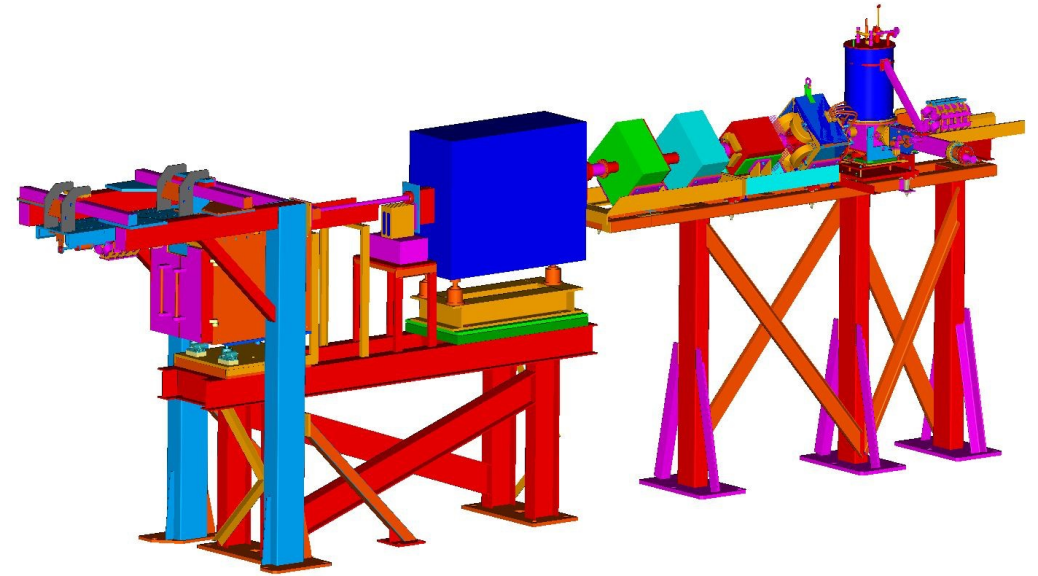
Compton and Moller polarimeters used extensively in 6 GeV era

Compton improvements during 6 GeV:

- Laser system ☑ IR to green
- Photon detector ☑ GSO crystal + threshold-less integrating DAQ

Møller improvements during 6 GeV:

- High field target magnet (3-4 T)



Improvements/modifications for 12 GeV

1. Compton chicane modifications (reduced vertical deflection)
2. Moller detectors repositioned (closer to beamline)
3. New, “cryogen-free” magnet for Møller target

Improvements to Hall A and C Polarimetry

- The polarimeters in both Halls A and C have steadily improved precision since the start of the JLab program
- Need for high precision primarily driven by PVES experiments (HAPPEX, PREX, CREX, Q-Weak)
- Precision of $dP/P \sim 1\%$ almost “routine”
- Future/ongoing polarimetry improvements driven by:
 - Unprecedented precision required by MOLLER and SOLID
 - Standardization between halls A and C
 - More reliable and robust operation

Experiment	Device	dP/P
HAPPEX-III	Compton (photon)	0.94%
PREX-1	Compton (photon)	1.1%
	Møller	1.2%
Q-Weak	Compton (electron)	0.6%
	Møller	0.85%
PREX-2	Møller	0.89%
CREX	Compton	0.52%
	Møller	0.85%
MOLLER		0.4%
SOLID		0.4%

0.44% combined!

Compton Polarimetry - HIPPOL

- **H**igh **P**recision **P**olarimetry capital project
- Ultimate goal: Compton polarimeters in Halls A and C with similar capabilities
 - Leverage available beamtime for systematic, functional tests
 - Use of same components allows for easier access to spares (fiber amplifiers, DAQ components, etc.)
- Scope:
 - Improve Hall C laser system to be similar to Hall A system (low gain cavity ☑ high gain cavity) ☑ components in hand
 - Larger electron detector for Hall C ☑ needed for 11 GeV operation
 - New electron detector in Hall A
 - Update Hall C electron detector DAQ (VETROC system) ☑ components in hand

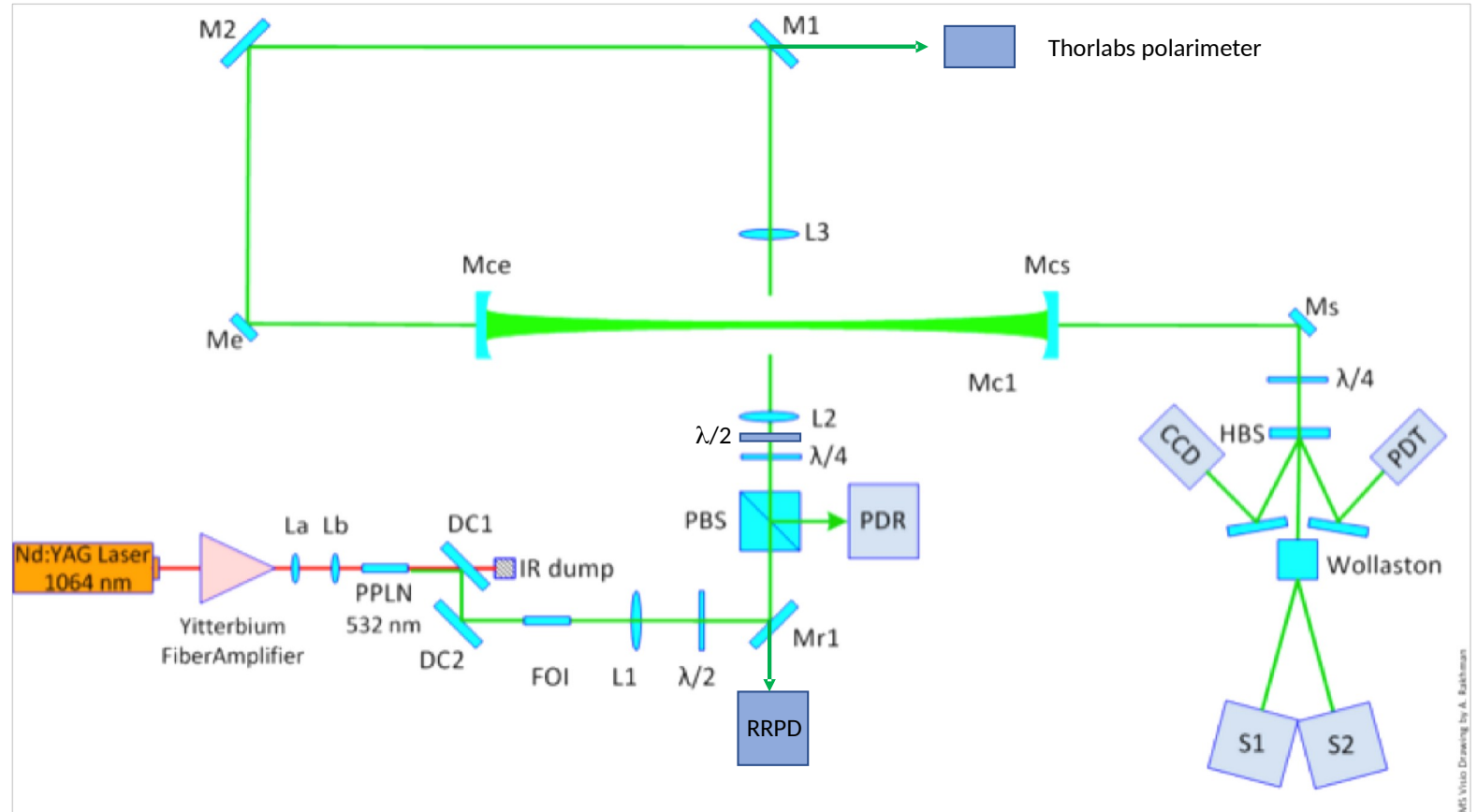
Hall A Laser System

Main components:

- Narrow linewidth 1064 nm seed laser
- Fiber amplifier (>5 W)
- PPLN doubling crystal
- High gain Fabry-Perot cavity
- Polarization manipulation/monitoring optics

Properties:

- 1 W laser power from doubling system
- Mirror reflectivity > 99.98%
- Cavity finesse $\geq 13,000$
- Stored power 2-10 kW



HIPPOL: seed laser, fiber amplifier, high reflectivity mirrors for Hall C

Other Laser system updates (not HIPPOL)

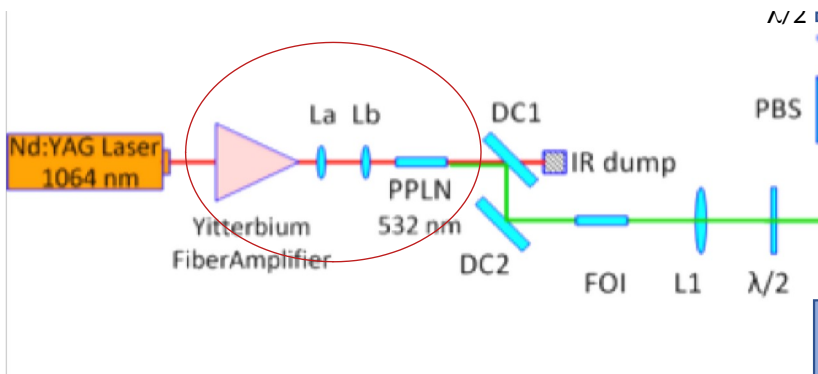
Locking electronics:

Hall A still using same custom cavity locking electronics from late 90's

- Locking electronics live upstairs in CH, VME/EPICS interface in hall
- CH electronics have already been replaced with spare modules
- VME modules in hall have had component failures in the last few years - repaired by Spectrometer Support

For reliable long term operation, need spares/backup system

- Hall C used commercial, FPGA-based locking system (Digilock) – may not be fast enough for high gain cavity
- Tests with Hall A cavity planned



Laser frequency doubling system

- Existing system uses separate fiber amplifier + PPLN crystal
- Replacing PPLN crystal requires significant alignment + downtime
- Combined-function amplifier-doubler will be tested for compatibility with high finesse cavity

Other Laser system updates (not HIPPOL)

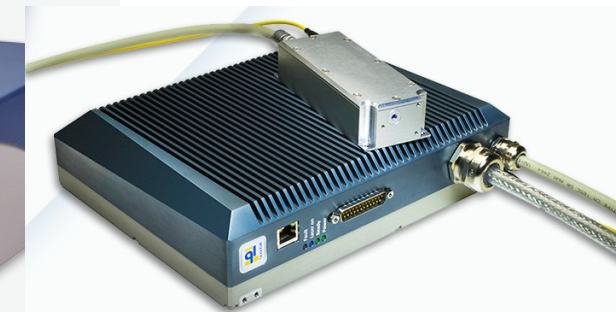
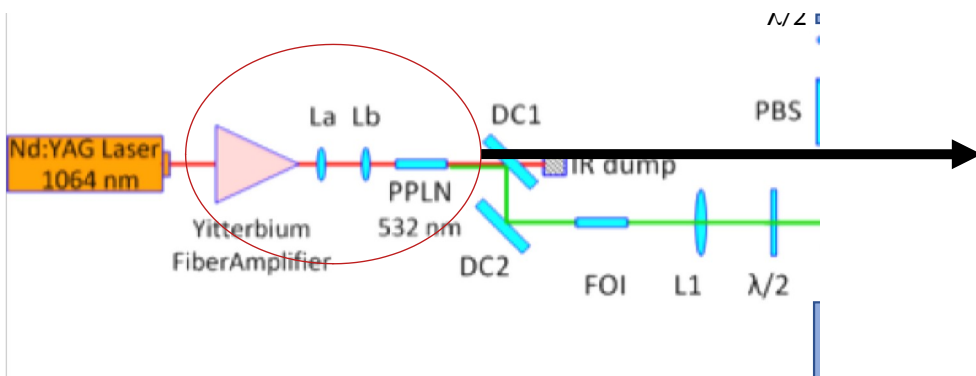
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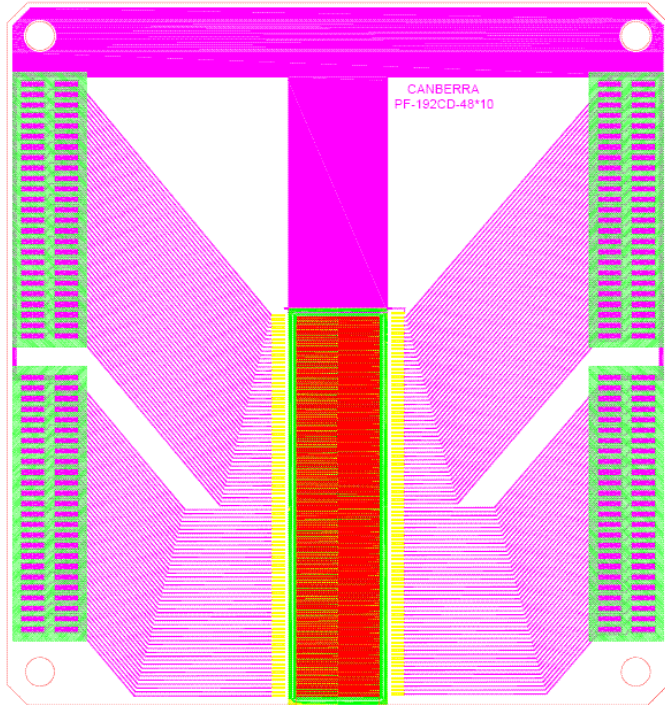
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Compton Electron Detector

Existing electron detectors

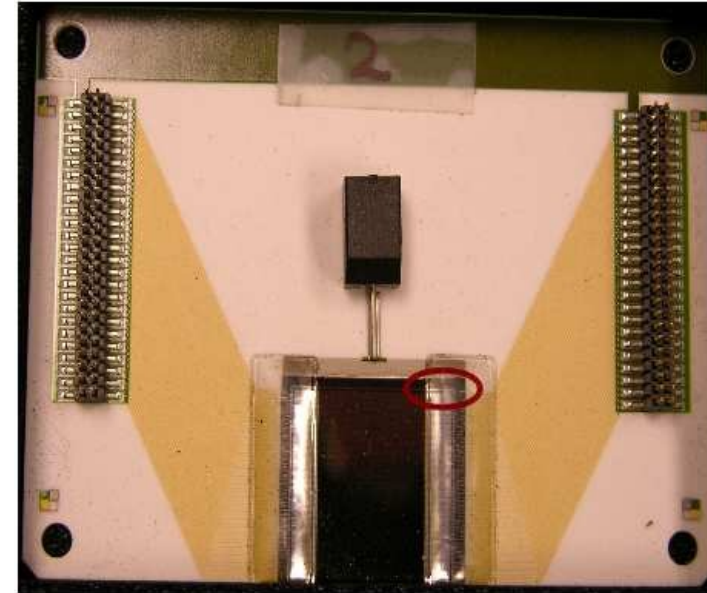


Hall A: silicon strip

→ 4.6 cm vertical coverage

☑ 192 strips, 240 μm pitch

→ *Suffers from small signal size/excess noise leading to low efficiency*



Hall C: diamond strips

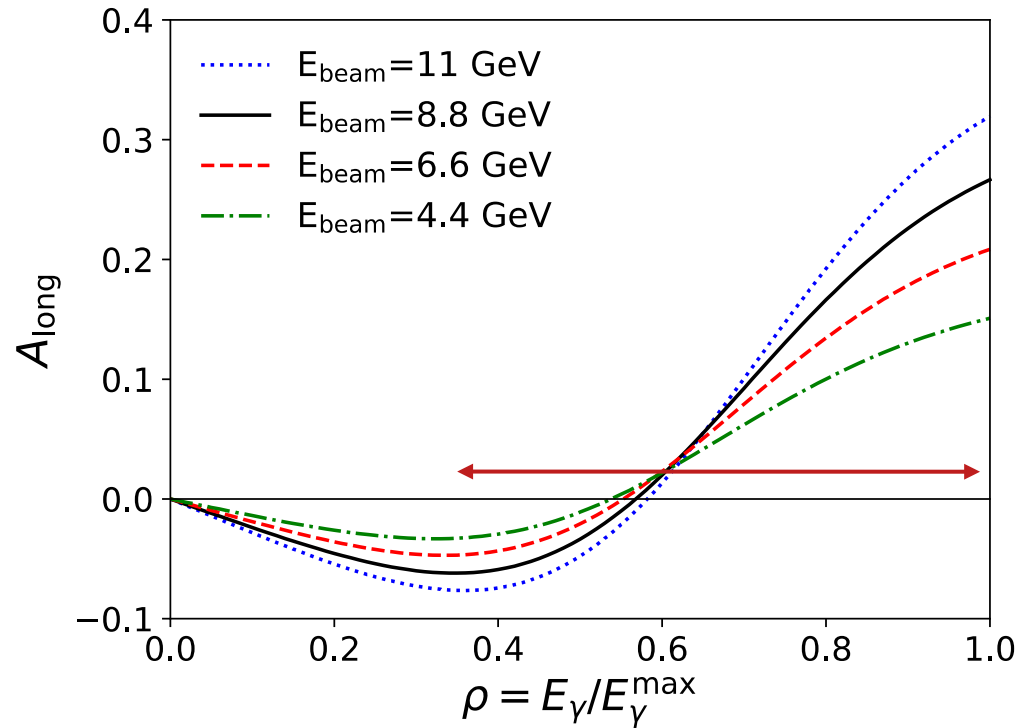
→ 2 cm vertical coverage

→ 96 strips, 200 μm pitch

→ *Undersized for ideal 11 GeV operation*

→ *Efficiency ok, but could be improved*

Detector Requirements



Needs to cover from Compton edge to at least zero crossing, asymmetry minimum preferred

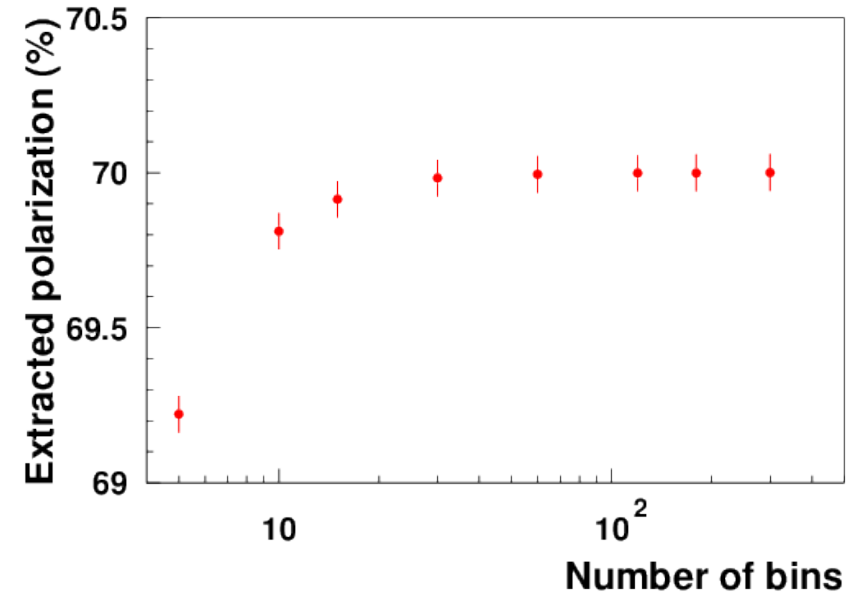
Hall A  5.75 cm

Hall C  > 2 cm



2 x 3 cm or 6 cm plane for Hall A

3 cm plane for Hall C



Need ~ 30 bins/strips between endpoint and zero-crossing to reliably extract polarization

A pitch of 245 μm will allow good performance down to 4.4 GeV in both Hall A and C

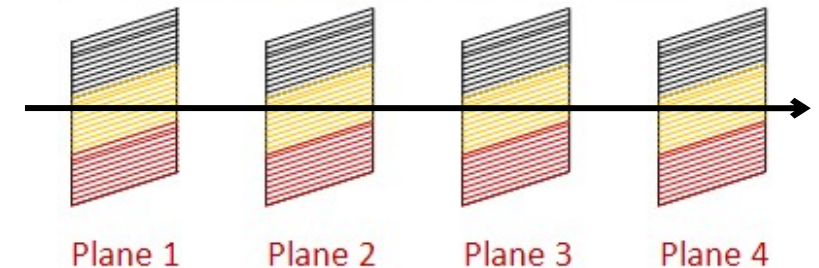
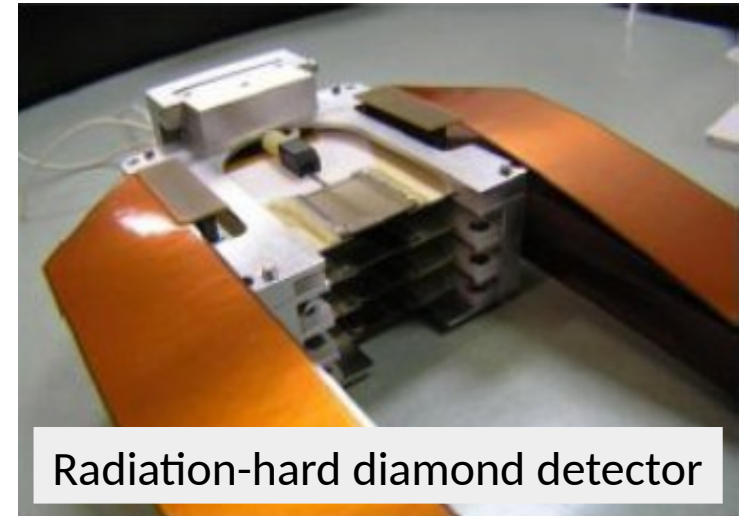
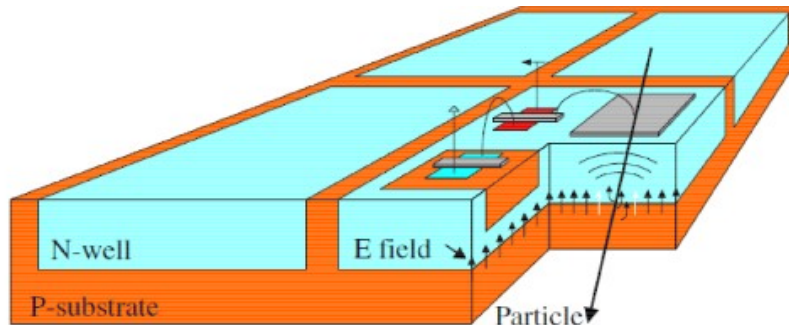
Electron Detector

HIPPOL project:

- Replace existing electron detectors in Halls A and C with new diamond strip detectors
- Optimize size and granularity for operation at 11 GeV while retaining low energy capabilities (perhaps slightly reduced precision at lowest energy)
- Improve on Hall C design with different approach to electronics – amplify signal on detector plane (rather than outside vacuum can)

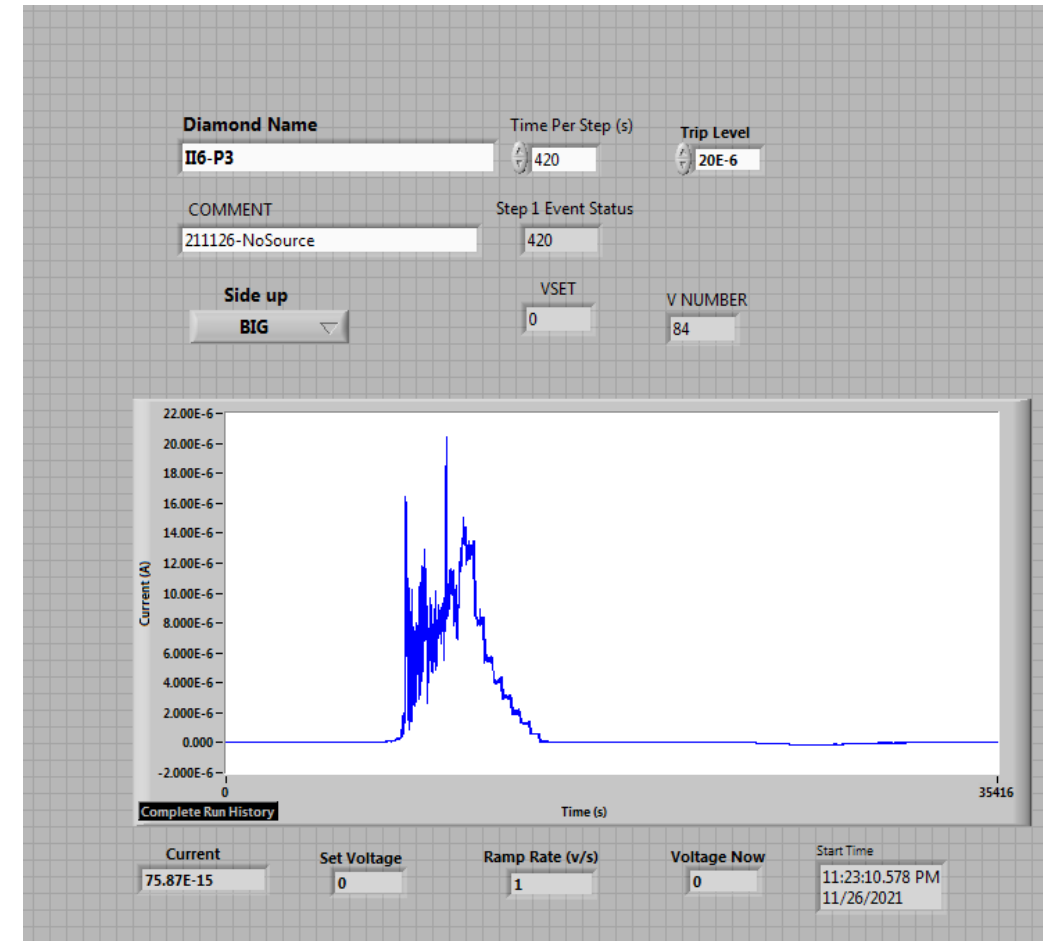
In parallel, U. Manitoba developing HVMAPS-based detector system (Hall A only)

☑ Prototyping and testing underway



Electron Detector

- Hall C diamond detector worked for Q-Weak but suffered from non-uniform, modest efficiency
 - Analogue pulses carried out of vacuum via flex cables
 - QWAD amplifier-discriminators just outside vacuum can
- New diamond detectors will use ASICs on detector board in place of QWADS outside of vacuum can
- Candidate ASICs under consideration
 - SAMPA (ALICE)
 - SALT (LHCb)
 - Calypso (Ohio State)
- Status
 - Test diamond planes obtained from II-VI, sent to Ohio State for characterization
 - High leakage currents (μA instead of pA) ✉ boards will be replaced
 - SALT/SAMPA board design complete



Tracking Detectors for Hall A Møller

Existing Hall A Møller detector system has only coarse information on event distribution at detectors

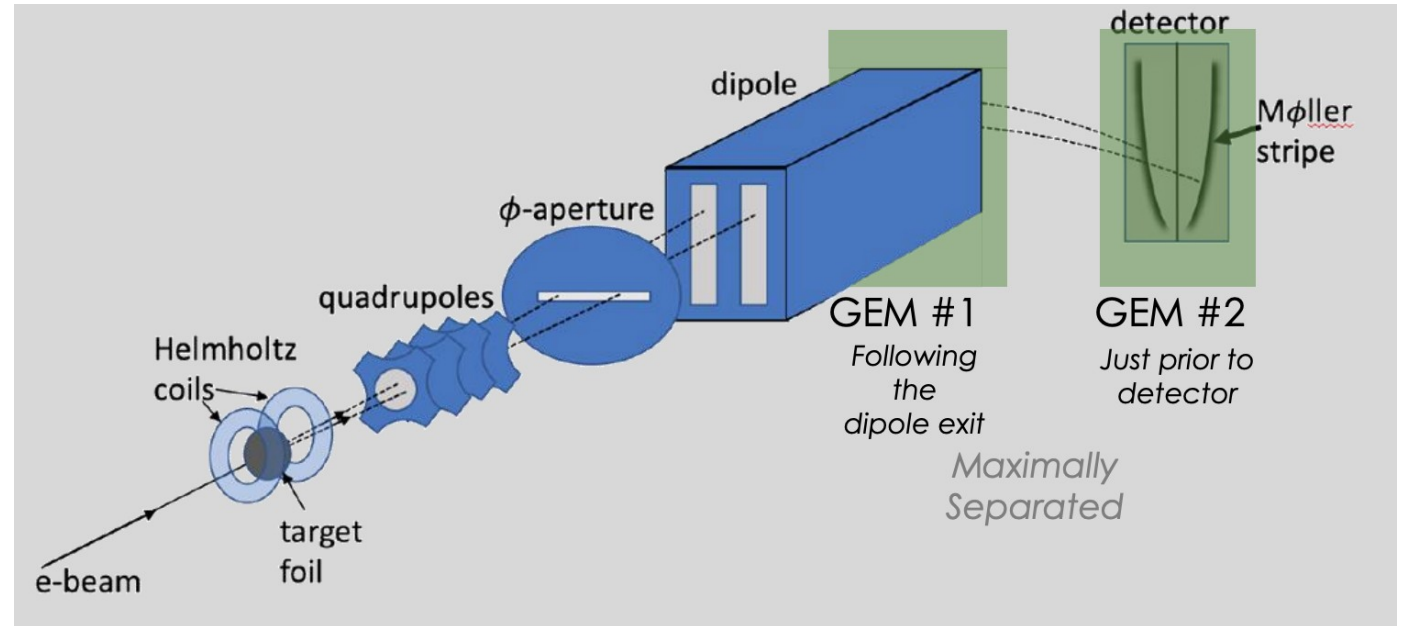
- Event distribution can provide information about spectrometer optics ✉ check Monte Carlo (analyzing power)
- Additional tracking detector would help reduce systematic error due to acceptance/optics
- Studies underway to see how tracking detector could help studies of radiative effects and Levchuk effect correction

GEMs will be used to provide this tracking information

- Funded through MOLLER NSF midscale project (detectors) and DOE project (mechanical supports)

Detector design nearing completion

- Preliminary design review in January
- Desirable to install early to allow use/checkout prior to MOLLER run



Møller GEM Design

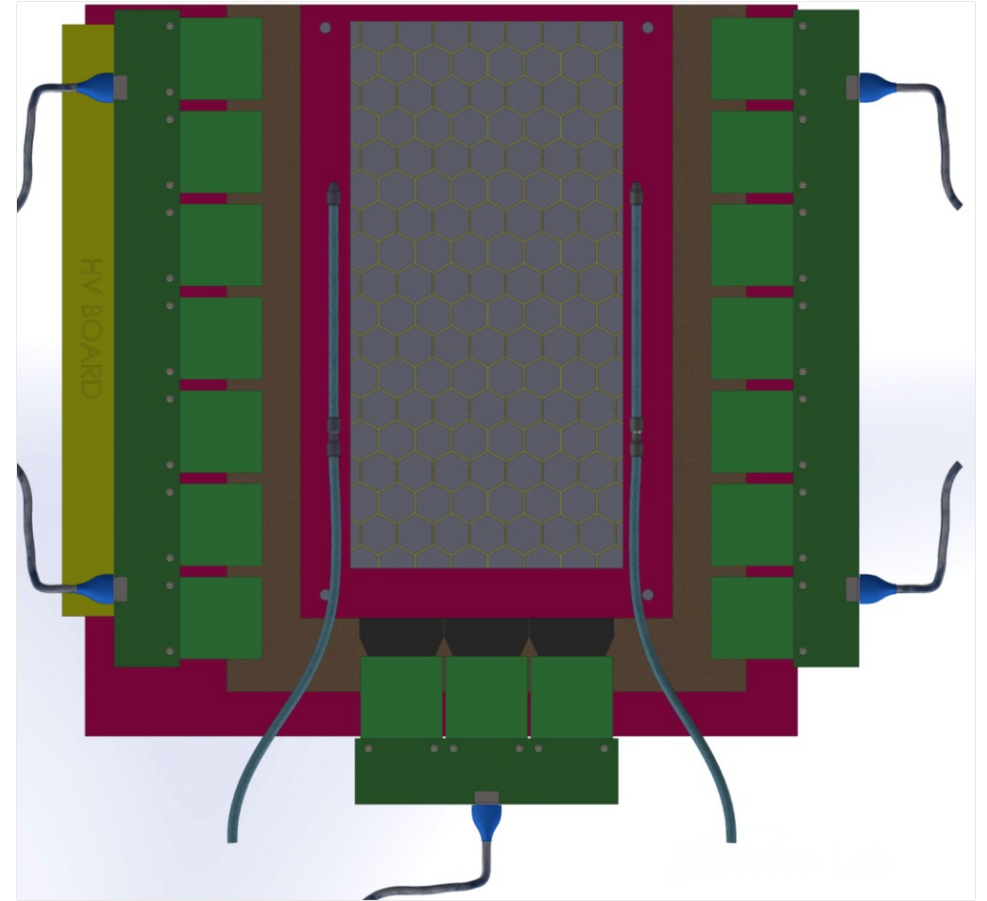
Møller polarimeter GEM detector included in Preliminary Design Review (Jan. 12-14, 2022) held for several MOLLER detectors

Committee recommended adding a 3rd GEM plane (current plan has only 2)

- In subsequent discussions, leaning towards having 4 planes total, with a pair in two locations

Other proposed changes/updates post-review:

- Reduce height of active area for better integration with beamline
- Optimize connectors and cable routing
- Start working with JLab designers to determine plan for detector supports



Moller Polarimeter GEM preliminary design,
Nilanga Liyanage (Uva)

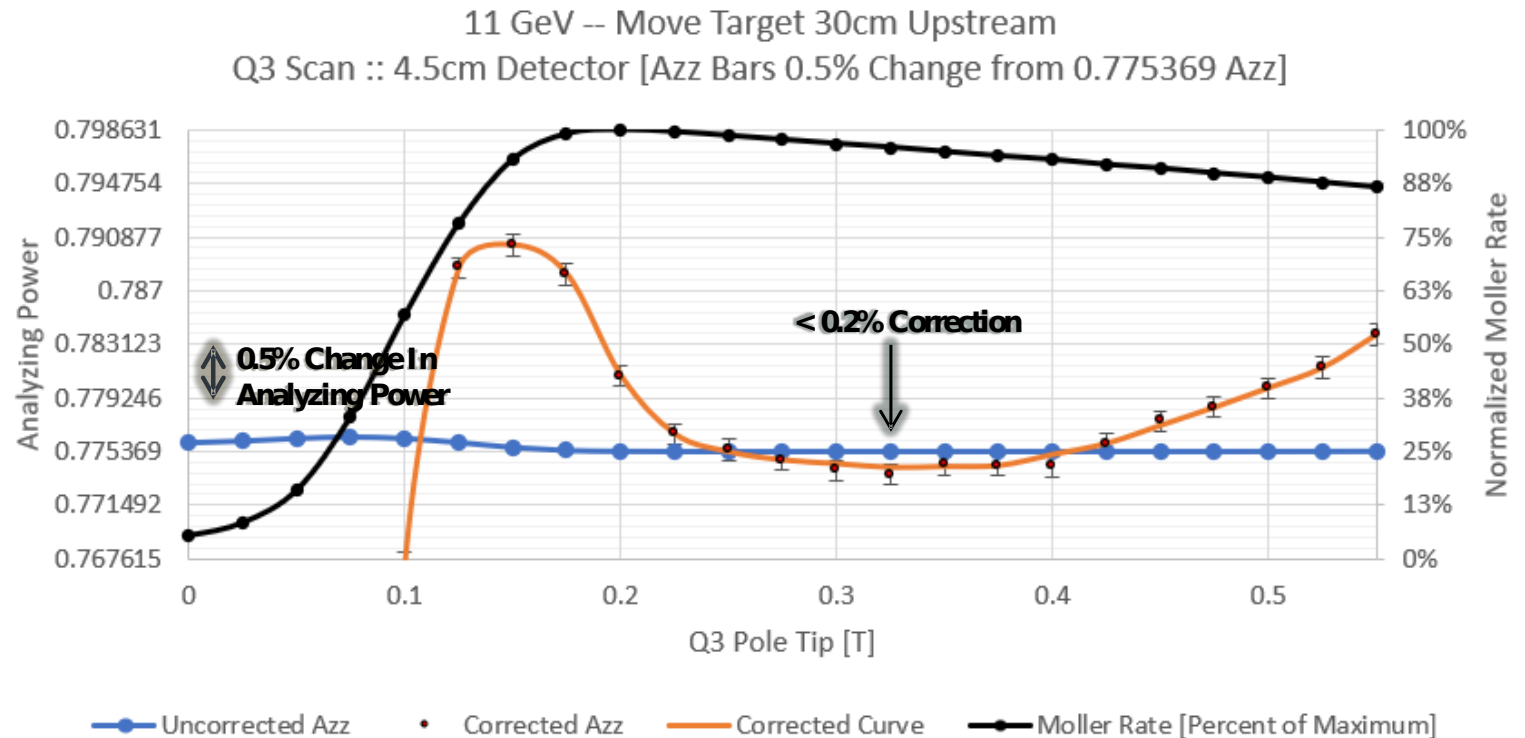
Møller Target Magnet – Position Shift

At 11 GeV, Møller quads not strong enough to totally eliminate impact of so-called Levchuk effect

→ Correction for Levchuk effect model-dependent and tightly coupled with detailed understanding of acceptance

Can be mitigated by shifting the Møller target/magnet upstream by ~30 cm

Target magnet shift included in MOLLER beamline design ✉ also being pursued in the near-term if time and resources allow



Eric King – Temple University

New Collimator for Moller detectors

In addition to target position shift, changes to acceptance also required to mitigate Levchuk effect

☑ Minimizing Levchuk effect requires vertical detector acceptance of 4.5 cm

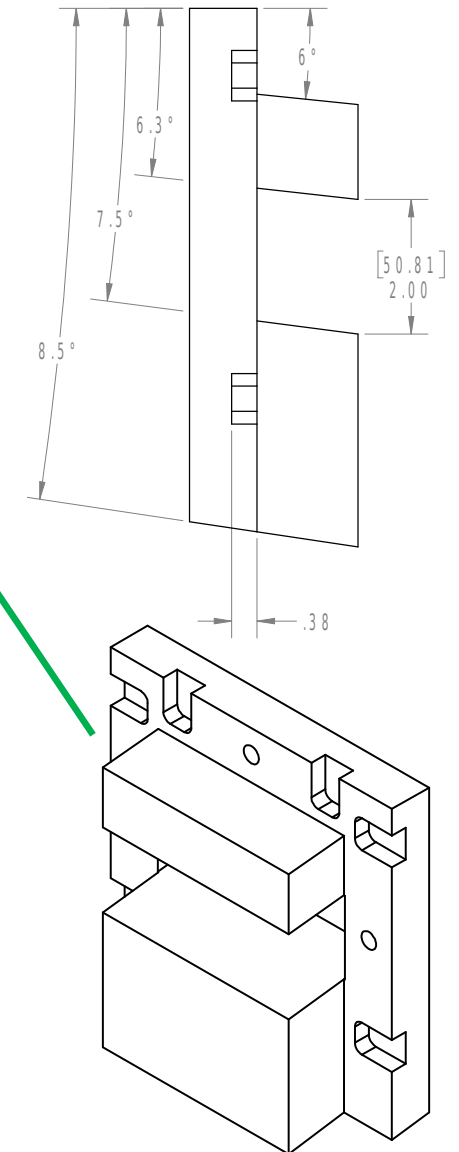
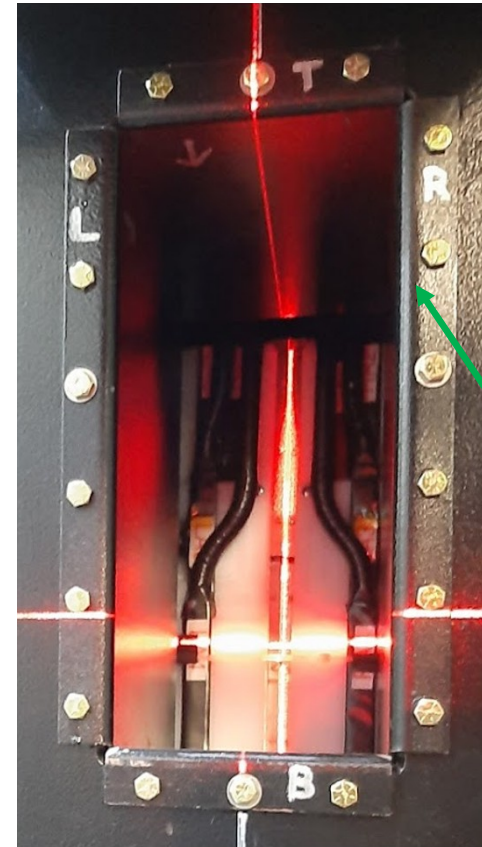
→ Full acceptance of Møller detectors is 30 cm – this can be reduced by turning off PMTs, but minimum coverage is ~7.5 cm

Rather than fabricating new (smaller) detectors and possibly impacting operation at lower energy, opted to design a small removable collimator

→ Collimator will mount on detector opening (using existing bolts)

→ Bottom detectors can be turned off, so no need to extend all the way to the bottom of the detector

Picture of window with frame



Collimator concept: Jim Fast, Don Jones, Eric King

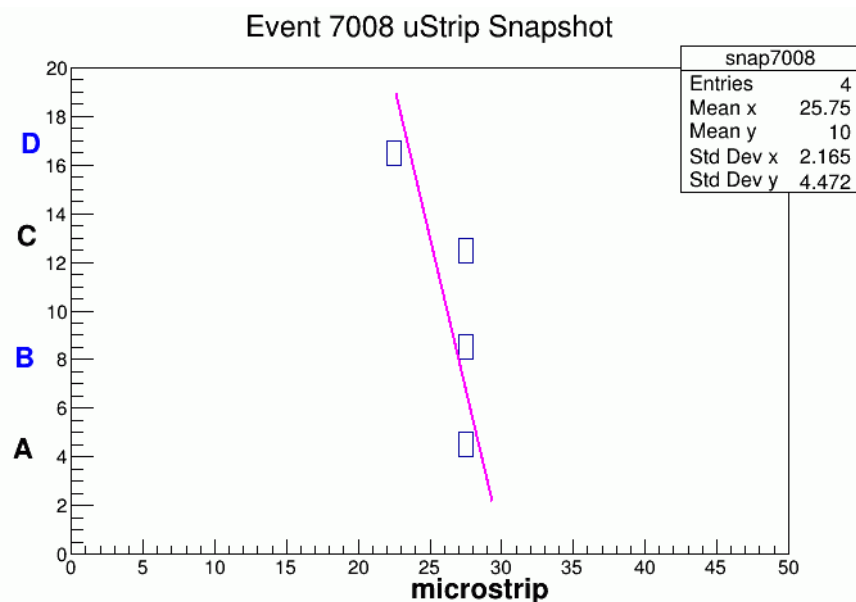
Summary

- Several ongoing/planned improvements to Hall A (and C) Compton polarimeters
 - HIPPOL capital project
 - Hall C Compton laser
 - Hall C electron detector DAQ
 - Hall A and C electron detectors
 - Additional improvements being pursued for Compton laser system
 - New locking electronics
 - Combined function amplifier and frequency doubler
- Hall A Møller
 - New tracking detectors to better constrain analyzing power and corrections
 - Target shift ~30 cm upstream
 - Acceptance defining collimator

Electron Detector DAQ

Electron detector readout requires processing “tracks” from multiple planes at high rates (order 100s of kHz)

- Hall A has transitioned to using the JLab VETROC modules (first use with beam during CREX)
- Hall C was using CAEN V1495 during Q-Weak
 - V1495 worked well, but can only handle full tracks at very limited rates
 - Required complicated pseudo-tracking based trigger with operation in scaler mode
- Hall C also moving to VETROC-based readout
 - Standardization across Hall A and C
 - Higher rate capabilities – more flexibility



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